AUTONOMOUS STUDIO · PRODUCTION SYSTEM DOCUMENTATION · MARCH 2026

FIELD GUIDE

Tutorial & Reference

A comprehensive walkthrough of the Autonomous Studio pipeline — from lore ingestion and knowledge graph construction through LoRA training, agent orchestration, and full episode generation. Uses the Dark Matter Universe as the working example throughout.

8	6	9	21,847
CHAPTERS	AGENTS	LORA ADAPTERS	VECTORS

AUTONOMOUS STUDIO · LOCAL PRODUCTION SYSTEM

4-NODE GPU CLUSTER · RTX A4000 / RTX 5090 / RTX 3060 / M1 MAX

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01

WELCOME BRIEFING

Mission Control

You are operating an autonomous AI production system — a virtual studio capable of generating complete TV and film content at series scale, with full continuity, consistent character voices, and cinematic visual output.

This dashboard is your mission control. Every panel reflects a live system component. Each section controls a real part of the pipeline — from the 4-node GPU cluster to the 6-agent LangGraph orchestration layer to the multimodal visual output pipeline.

SYSTEM CAPABILITIES

WHAT THE SYSTEM DOES

Vector Memory	Semantic retrieval across 21,847 embedded lore chunks via Qdrant
Knowledge Graph	Neo4j continuity constraint checking — 1,203 nodes, 3,847 relationships
LoRA Adapters	9 fine-tuned adapters spanning universe, series, and character voice
6-Agent Pipeline	Showrunner → World Builder → Head Writer → Scene Writer → Story Editor → Researcher
Multimodal Output	Script, character images (FLUX LoRA), and motion video (WAN2.2) from a single brief

THE 5-STAGE WORKFLOW

1 INGEST LORE → 2 EMBED + GRAPH → 3 TRAIN LORAS → 4 CONFIGURE AGENTS → 5 GENERATE

> ⋆ This guide uses the Dark Matter Universe — a fully loaded example world — throughout. All characters, factions, and locations referenced here are real entities in the knowledge graph, accessible in the Lore Library section of the dashboard.

WORLD REFERENCE

02 The Dark Matter Universe

The Dark Matter Universe is a multi-series science fiction world modelled after productions like *The Expanse* and *Andor* — morally complex, politically driven, with a rich institutional history. It is the working example universe for all tutorials.

342	48	127	412	2	21,847	1,203
CHARACTERS	FACTIONS	LOCATIONS	EVENTS	ACTIVE SERIES	VECTORS	GRAPH NODES

CHARACTER DOSSIERS — MAIN CAST

KAEL THERON

FIELD COMMANDER → DEFECTOR · IRON CONCLAVE (FORMER)

STATUS ALIVE
FIRST SEEN S01E01
VOICE LORA kael_theron_voice (r=16)

A former enforcer for the Iron Conclave who defected after discovering the Vault-7 conspiracy. Tactically brilliant but morally compromised by his past service. Speaks in clipped, precise sentences. Rarely uses contractions. Favours understatement in moments of high tension.

Tactically brilliant | Morally compromised | Clipped speech | Rarely uses contractions | Understatement under pressure

> *"Six months. You knew. The whole time."*

SERA VANCE

INTELLIGENCE OPERATIVE · INDEPENDENT

STATUS ALIVE
FIRST SEEN S01E01
VOICE LORA sera_vance_voice (r=16)

Operative with no permanent allegiance. Allies with Kael out of mutual necessity. Their tension is unspoken and persistent. She plays the long game — always has an exit prepared. Ruthlessly pragmatic, dry wit, distrusts ideology.

Ruthlessly pragmatic | Dry wit | Long-game thinker | Distrusts ideology

> *"I didn't save you. I preserved a useful asset. There's a difference."*

DIRECTOR MAREN

HEAD OF THE IRON CONCLAVE · ANTAGONIST

Kael's former superior and mentor. Authorised the Vault-7 programme from a cold calculation about survival. Never raises her voice. Genuinely sorrowful about Kael's defection.

> *"I gave you every opportunity to understand. You chose sentiment."*

THE ARCHIVIST

INFORMATION BROKER · NEUTRAL

Entity of unknown origin. Provides intelligence to Kael on a strictly transactional basis. Speaks in precise riddles. Never lies; rarely tells the full truth.

> *"The vault was sealed three centuries before your organisation existed. Ask yourself who built the lock."*

MAJOR FACTIONS

IRON CONCLAVE

ACTIVE

The dominant political-military body. Controls dark-field technology that underpins interstellar transit. Enforces "civilisational stability."

THE DRIFT COLLECTIVE

ACTIVE

Loose federation of outer-rim stations that rejected Conclave governance. Resourceful and decentralised. Kael's primary allies post-defection.

THE VAULT PROGRAMME

CLASSIFIED

The conspiracy at the centre of Series 01. A long-running Conclave project to weaponise dark-field technology for population management.

END-TO-END WORKFLOW

03 Full Production Pipeline

Every episode passes through six stages. Each stage feeds the next — raw lore becomes structured memory, memory enables accurate generation, generation produces both text and visuals in parallel.

01

LORE INGESTION

Drop raw source material — PDFs, scripts, markdown notes — into the Lore Library ingestion pipeline. The system parses, chunks, and classifies entities automatically.

Example: darkmatter_S02_bible.pdf → 2,847 chunks · 14 entity types classified

→ Text chunks
Entity candidates

02

EMBED + INDEX

nomic-embed-text-v1.5 encodes every chunk into 768-dimensional vectors. Vectors committed to Qdrant. Relationships extracted and written to Neo4j.

21,847 total vectors · 8 collections · Cosine similarity · HNSW m=16

→ Qdrant collections
Neo4j graph

03

LORA TRAINING

Unsloth QLoRA trains universe, series, and character voice adapters on Qwen2.5-14B-Instruct. Training data auto-generated from lore entities.

kael_theron_voice: 1,402 pairs · r=16 · 47 min · final loss: 0.724

→ Adapter files
(.safetensors)

04

ADAPTER LOADING

vLLM multi-LoRA loads the trained adapters. LiteLLM routes agent requests to the correct model+adapter combination via alias resolution.

7 adapters loaded · universe_voice merged · ranks r=16 to r=64

→ vLLM ready
LiteLLM routing

05

AGENT PIPELINE

LangGraph orchestrates 6 specialised agents. Each agent RAG-queries Qdrant before generating, then Neo4j validates the output for continuity.

Showrunner → Head Writer → Scene Writer × 12 → Story Editor

→ Script
Beat sheet

06

VISUAL PIPELINE

ComfyUI renders character images and scene illustrations using FLUX LoRAs. WAN2.2 generates motion video clips for key

→ Images
Video frames

scenes.

darkmatter_style (0.55) + kael_identity (0.85) · 28 steps · 1024×576

EPISODE TRACE — DARK MATTER S02E04

GENERATION PIPELINE — EPISODE TRACE

✓	INGESTION	darkmatter_S02_bible.pdf	→	2,847 chunks → Qdrant
✓	EMBEDDING	2,847 chunks · nomic-embed-text	→	+2,847 vectors committed
✓	GRAPH	Entity extraction pass	→	47 new relationships → Neo4j
✓	TRAINING	kael_theron_voice (1,402 pairs)	→	Adapter ready · loss 0.724
✓	SHOWRUNNER	Brief: S02E04 "The Vault Antechamber"	→	12 scene outlines
◌	SCENE WRITER	Beat 12: Kael confronts Maren	→	2,847 tokens generated
○	VISUAL	Scene 12 + LoRA stack	→	6 frames queued · ComfyUI

HARDWARE ARCHITECTURE

04 Your 4-Node GPU Cluster

The cluster is a heterogeneous network of four machines, each specialised for a different role. LiteLLM on the hub acts as a unified proxy — all agents address LiteLLM, which routes to the correct node and model via alias resolution.

● UBUNTU HUB

RTX A4000 · 16GB VRAM · 192.168.1.10

VRAM 70%

Qdrant :6333 | Neo4j :7474 | LiteLLM :4000 | n8n :5678

Always-on infrastructure. Low VRAM, high availability. Runs the memory layer and all routing.

◌ WIN WORKSTATION

RTX 5090 · 32GB VRAM · 192.168.1.42

VRAM 78%

vLLM :8000 | ComfyUI :8188 | Unsloth :7860

32GB VRAM fits the full 14B model Q4 + 3 LoRA adapters + KV cache simultaneously. The production workhorse.

● LAPTOP

RTX 3060 · 12GB VRAM · 192.168.1.55

VRAM 35%

Ollama :11434 | vLLM (overflow)

Handles lighter 7B tasks via Ollama. Overflow capacity when the 5090 is training.

● MACBOOK PRO

M1 Max · 32GB Unified · 192.168.1.71

VRAM 26%

LangGraph :3001 | MLX | n8n

LangGraph agent development and rapid iteration. MLX runs small models for testing.

◈ **LiteLLM alias convention:** Agents always reference models by alias (`qwen2.5-14b-vllm`, not raw endpoints). This means you can migrate a model to a different node without touching any agent configuration — only the LiteLLM alias mapping changes.

SERVICE REGISTRY

ALL RUNNING SERVICES

SERVICE	NODE	PORT	VERSION
LiteLLM Proxy	Ubuntu Hub	4000	v1.42.1
Qdrant	Ubuntu Hub	6333	v1.13.2
Neo4j Community	Ubuntu Hub	7474	v5.25.0

SERVICE	NODE	PORT	VERSION
n8n	Ubuntu Hub	5678	v1.68.0
vLLM (primary)	Win Workstation	8000	v0.8.1
ComfyUI	Win Workstation	8188	latest
Unsloth Studio	Win Workstation	7860	v2026.03
Ollama	Laptop	11434	v0.6.0

05

QDRANT + NEO4J

The Memory Layer

The memory layer has two complementary systems working in tandem. Qdrant handles semantic similarity searches. Neo4j handles structural relationship traversal and continuity constraint checking. Agents use both on every scene generation.

QDRANT — VECTOR COLLECTIONS

QDRANT · 192.168.1.10:6333 · 21,847 TOTAL VECTORS

COLLECTION	VECTORS	PURPOSE
universe_foundation	8,412	Core world rules, physics, timeline
darkmatter_episodic	4,231	Scene-level episodic memory
darkmatter_lore	2,847	Series-specific lore and context
darkmatter_dialogue	1,923	Canonical speech pattern examples
conclave_wars_s01	1,204	Series 2 lore (indexing)
characters_voice	892	Character voice profiles

EXAMPLE RAG QUERY

QDRANT SIMILARITY SEARCH · DARKMATTER_LORE + UNIVERSE_FOUNDATION

Query: "What does Kael know about the dark-field inhibitor?"

0.934	*"Kael had seen the inhibitor specifications only once, in the Conclave's restricted archive..."*	[S01E03]
0.901	*"The inhibitor was never meant for civilian application—"*	[S01E05]
0.878	*"Dark-field inhibition requires neural isolation at the cortical binding layer..."*	[universe]

Latency: 12ms · Top-k: 7 · Distance: Cosine

NEO4J — KNOWLEDGE GRAPH

GRAPH STATISTICS

Total Nodes	1,203
Relationships	3,847
Characters	342
Factions	48
Locations	127
Events	412

RECENT CONSTRAINT CHECKS

✓ PASS · Kael Theron · status: ALIVE · S02E06

✓ PASS · Location 'Vault-7' · established S01E01

✗ FAIL · "Kel Aran" referenced ALIVE — status: DEAD (S01E02)

◈ The Lore Layer hybrid query tester lets you run both a Qdrant semantic search and a Neo4j Cypher query simultaneously, with results merged and ranked. Use this to verify that a piece of lore is both semantically retrievable and structurally consistent before writing new content.

06

LORA ARCHITECTURE & COMPOSITION

Adapter Vault

LoRA adapters are the system's institutional memory at the model level. Where Qdrant stores *what happened*, LoRAs encode *how the universe speaks* — the specific narrative voice, vocabulary, and stylistic fingerprint of the fiction.

THE THREE-TIER HIERARCHY

T1 UNIVERSE

universe_voice — r=64 · MERGED · Always active

Trained on all universe-level world-building documents. The foundational voice of the entire Dark Matter Universe.

Every other adapter stacks on top of this. Because it is merged into the base model weights at deployment time, it adds zero VRAM overhead during inference. All agents inherit this voice.

T2 SERIES

darkmatter_s01 · darkmatter_s02 — r=32 · weight: 1.0

Series-specific lore, recurring locations, faction dynamics, season arc knowledge. Swapped per generation job.

Only one T2 LLM adapter should be active at a time. Applying both series adapters simultaneously causes interference and degrades output coherence.

T3 CHARACTER

kael_theron_voice · sera_vance_voice — r=16 · weight: 0.7 max

Individual character voice adapters. Applied at reduced weight to preserve series tone.

Never combine two T3 LLM adapters in a single stack — interference risk. Stack T1 (merged) + T2 (1.0) + one T3 (≤0.7) maximum per agent run.

EXAMPLE COMPOSITION — KAEL SCENE · RTX 5090 VRAM BUDGET

LORA STACK · SCENE WRITER · KAEL POV

T1 · universe_voice · r=64
MERGED — always active, zero overhead

T2 · darkmatter_s02 · r=32 · ×1.0
Series lore, tone, continuity. Full weight.

T3 · kael_theron_voice · r=16 · ×0.7

VRAM BUDGET · RTX 5090 (32GB)

Qwen2.5-14B Q4	6.80 GB
darkmatter_s02 (r=32)	0.17 GB
kael_voice (r=16)	0.09 GB
KV Cache	2.00 GB
TOTAL IN USE	9.06 GB ✓ Fits

Character voice subordinate to series tone.

! Do not combine two T3 LLM adapters in a single stack. Interference between competing character voices degrades output quality significantly. Maximum stack: T1 (merged) + T2 (1.0) + one T3 (≤0.7).

FULL ADAPTER REGISTRY

ALL REGISTERED ADAPTERS · AUTONOMOUS STUDIO

NAME	TYPE	BASE	RANK	TIER	LOSS
universe_voice	LLM	Qwen2.5-14B	r=64	T1	0.681
darkmatter_s01	LLM	Qwen2.5-14B	r=32	T2	0.712
darkmatter_s02	LLM	Qwen2.5-14B	r=32	T2	0.724
kael_theron_voice	LLM	Qwen2.5-14B	r=16	T3	0.734
sera_vance_voice	LLM	Qwen2.5-14B	r=16	T3	0.748
darkmatter_style	FLUX	FLUX.1-dev	r=32	T2	—
kael_identity	FLUX	FLUX.1-dev	r=16	T3	—
sera_identity	FLUX	FLUX.1-dev	r=16	T3	—
darkmatter_video	WAN2.2	Wan2.2-T2V-14B	r=16	T2	TRAINING

AGENT ORCHESTRATION REFERENCE

07 The 6-Agent Pipeline

The pipeline is a LangGraph directed graph. The Showrunner is the supervisor node — it receives the brief, delegates to writers, collects output, and coordinates the editorial pass. Each agent has its own model, LoRA stack, and memory configuration.

AGENT ROSTER

SHOWRUNNER

Qwen2.5-32B · High capability orchestrator

universe_voice

Orchestrator and supervisor. Receives the episode brief, breaks it into a beat sheet, assigns scenes to the Head Writer, manages continuity at episode level, and coordinates the editorial pass.

WORLD BUILDER

Qwen2.5-14B · Long context

universe_voice | darkmatter_s02

Expands series bible entries, creates new locations and faction lore on demand. Invoked when the Showrunner requires new canonical world-building not present in existing lore.

HEAD WRITER

Qwen2.5-14B · Structured output

universe_voice | darkmatter_s02

Produces beat sheets and scene outlines from the Showrunner's episode brief. Specifies story function, primary characters, setting, and character state (in/out) for each scene.

SCENE WRITER

Qwen2.5-14B · Creative generation

darkmatter_s02 | kael_theron_voice

Generates full scene scripts. RAG-queries Qdrant before each scene (7 chunks average). Receives Neo4j constraint feedback inline. LoRA stack swaps per character focus.

STORY EDITOR

Qwen2.5-7B · Fast evaluator

universe_voice

Reviews the full episode draft. Checks: character consistency, continuity violations, pacing. Returns pass/fail with specific inline notes. Triggers rewrites for failed scenes.

RESEARCHER

Qwen2.5-7B · RAG specialist

No LoRA — pure retrieval

Lore retrieval specialist. Queries Qdrant and Neo4j on behalf of other agents. Formats and ranks results. Does not generate narrative content — only retrieves and structures existing lore.

INT. VAULT ANTECHAMBER — NIGHT

The antechamber is cold. Kael's breath fogs in the emergency lighting. His hand stays near his holster — not touching it. Not yet.

MAREN stands at the far console, her back to him. She knew he was coming.

KAEL

Six months.

MAREN

(not turning)
Longer than that.

KAEL

You knew. The whole time.

A beat. She turns. There's no defiance in her face — only something that might be called regret, if the Iron Conclave had ever trained its directors to show it.

MAREN

I gave you every opportunity to understand.
You chose sentiment.

★ Kael's dialogue — short sentences, no contractions, maximum information density — is the kael_theron_voice adapter (r=16, weight 0.7) at work, trained on 1,402 canonical dialogue pairs extracted from the series bible.

PRODUCTION WALKTHROUGH

08 Running a Generation Job

This walkthrough generates a complete episode of Dark Matter — Series 02, Episode 04. Each step shows what happens at the system level during a production run.

6-STAGE GENERATION WALKTHROUGH

STEP 01 — BRIEF SUBMISSION

Navigate to Generation → New Job. Select: Series: Dark Matter, Season: 02, Episode: 04.

Enter the one-line brief:

```
"S02E04 — Kael confronts Maren in the Vault antechamber. The Conclave's
secret is revealed."
```

STEP 02 — SHOWRUNNER

The Showrunner receives the brief and produces a 12-scene beat sheet. Each beat specifies: story function, primary characters, setting, and character state going in/out.

Output: 12 beats · ~800 tokens · 8.2s · Qwen2.5-32B + universe_voice

STEP 03 — HEAD WRITER

The Head Writer expands each beat into a full scene outline — setting, action, dialogue purpose, character emotional state in/out. Serves as the blueprint for the Scene Writer.

Output: 12 outlines · ~3,200 tokens · 22s

STEP 04 — SCENE WRITER (×12)

The Scene Writer generates each scene in sequence. Before writing each scene, it executes:

1. RAG query to Qdrant (avg 7 chunks, 12ms) — retrieves relevant lore context
2. Neo4j constraint check — validates all referenced entities are alive/accessible
3. Generates scene with LoRA stack: darkmatter_s02 (1.0) + kael_theron_voice (0.7)

Output: 12 full scenes · ~28,000 tokens · ~8 min total

STEP 05 — STORY EDITOR

The Story Editor reviews the complete draft. It checks character consistency, continuity violations, and pacing. Issues inline correction notes. Scenes that fail are automatically requeued for rewrite with the constraint injected into the prompt.

Output: Pass 94% · 2 constraint flags · Revision notes · Qwen2.5-7B

STEP 06 — VISUAL PIPELINE

ComfyUI renders key frames for flagged and pivotal scenes using the FLUX LoRA stack. WAN2.2 generates motion clips for specified scenes if video is requested.

Output: 24 frames · 6 motion clips · darkmatter_style (0.55) + kael_identity (0.85)

EPISODE OUTPUT SUMMARY — DM S02E04

12 SCENES	~32K TOKENS	84 RAG QUERIES	36 GRAPH CHECKS
2 VIOLATIONS	94% CONSISTENCY	24 IMAGES GEN	~12 min RUNTIME

◈ The 2 continuity violations are automatically requeued: the Scene Writer regenerates those specific scenes with the constraint note injected into the prompt, informed by the exact Neo4j path that failed. No human intervention required.

WHERE TO START

RECOMMENDED FIRST STEPS

1. Browse the Lore Library — explore Kael Theron's entity detail
2. Open the Adapter Vault — inspect the LoRA registry and composition planner
3. Go to Training — review the active job and live loss curve

QUICK REFERENCE — KEYBOARD SHORTCUTS

Sidebar — hover to expand, auto-collapses

Agent Config — click [CONFIGURE] in Agents page

Query Tester — Lore Layer → bottom panel

4. Navigate to Generation — launch a run for DM S02E05

Loss Chart — Training page, live via recharts